

Week 8 – Templates, Component Extension & Dispatcher

Overview

This week focused on deeper AEM development concepts including static and editable templates, overlay vs override mechanisms, and dispatcher architecture. Practical assignments were centered around extending existing AEM Core Components and customizing their dialogs and HTML structure without breaking inheritance rules.

Implementation Details

1. Static & Editable Templates

- Studied template types in AEM:
 - **Static Templates** – defined in /apps and controlled by developers.
 - **Editable Templates** – defined in /conf and configurable by authors.
- Learned template structure including:
 - Template definition
 - Structure
 - Policies
 - Initial content
- Created pages using editable templates and analyzed how layout containers control allowed components.

2. Overlay vs Override

- Learned the difference between overlay and override in AEM development.
- **Overlay** – copying component/dialog structure from /libs to /apps to customize behavior.

- **Override** – replacing component functionality through inheritance using `sling:resourceSuperType`.
- Studied folder hierarchy (`/libs`, `/apps`) and best practices for extending components.

3. Dispatcher Architecture

- Studied the role of **Dispatcher** as AEM's caching and load-balancing layer.
- Key concepts learned:
 - Web server integration
 - Dispatcher filters and request rules
 - Cache management
 - Header handling
- Understood how dispatcher improves performance by caching responses and filtering requests.

4. Core Component Extension Assignments

Teaser Component (v1) Extension

- Created a component using Teaser v1.
- Added a new dialog tab **Styles** between **Text** and **Link & Actions**.
- Added custom fields under the Styles tab.

Carousel Component Extension

- Extended carousel component.
- Removed all fields from the **Accessibility tab**.
- Added new **ID text field**.
- Implemented without using `sling:hideResource`.

Button Component (v1) Extension

- Removed `fieldDescription` from the **Text field** under the Properties tab.
- Added new **ID field** under the Accessibility tab.

- Ensured the ID field appeared as the first field in that tab.

Button Component (v2) Extension

- Modified the component rendering logic.
- Appended custom class **custom-btn** to the icon span tag in HTL.

Testing & Results

- Extended components rendered correctly in author environment.
- Dialog modifications appeared correctly in page properties.
- Custom fields captured data successfully.
- HTML modifications reflected correctly on page rendering.
- Component inheritance maintained core functionality.

Challenges & Solutions

- Understanding overlay vs override structure initially caused confusion - resolved through testing /libs and /apps hierarchies.
- Dialog restructuring required careful node placement - corrected using CRX/DE.
- Ensuring correct HTL modification without breaking component behavior.
- Dispatcher architecture concepts required additional documentation study.

Next Steps

- Work on advanced dispatcher configurations.
- Explore caching strategies.
- Implement custom Sling Models for extended components.
- Improve component styling using client libraries.